Platen Superheater Transient Simulation — Complete User Guide

Application: Unit №4 Platen Superheater Transient Simulator (Android)

Platform: Android 8.0+ (API 26+)

Property engine: IAPWS-IF97 (Regions 1, 2, 4) — vendored `com.hummeling.if97` v2.1.0

Version: 1.0.0

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1. What the application does

This is an **engineering simulation tool**, not a static calculator. It computes how the **outlet steam temperature of the platen superheater** in a 325 MW natural-circulation drum boiler (Unit №4) responds **over time** to:

#	Driver
1	Steam mass flow / load changes
2	Spray-water flow changes at the platen inlet
3	Burner start/stop events
4	Furnace heat-input changes
5	Steam inlet pressure changes
6	Steam inlet temperature changes
7	Property changes with P and T (density ρ, cp, μ, k)
8	Internal convection coefficient h_i changing with flow and properties
9	Reynolds number changes with load
10	Prandtl number changes with steam state
11	Nusselt number changes with Re and Pr
12	Dynamic overall heat-transfer coefficient U
13	Tube-metal thermal inertia (the dominant delay source)
14	Steam thermal capacity / residence time
15	Spray injection cooling dynamics

The causal chain is physical, top to bottom:

```
P, T → IF97 properties (ρ, μ, cp, k)
     → velocity → Re → Pr → Nu → h_i → U
     → Q (per segment)
     → metal energy balance → steam energy balance
     → steam outlet T (with spray enthalpy + burner heat entering dynamically)
```

U is a calculated, load-dependent output — never a hard-coded constant.

Plant reference data (locked, Unit №4)

Parameter	Value
Boiler capacity	325 MW
Type	Natural circulation, drum-type
Burners	24 × 40 MW, 3 elevations
Full-load steam flow	1000–1040 t/h
Sliding pressure range	10–167 bar
Sliding steam temperature	~200–540 °C
Parallel panels	43
Tubes per panel	4 (172 total)
Passes per panel	8, pass length 6 m
Tube OD / wall / ID	45 / 8 / 29 mm
Inner (steam-side) area	~783 m²
Outer (furnace-side) area	~1216 m²
Header feed	Dual mid-point (panels 1–21 / 22–43)

Sections in flow order (this is the order steam actually flows, and the order the model uses):

#	Section	Length	Material
1	Inlet casing	0.5 m	12Cr1MoV
2	Lower radiant	15.0 m	SA-213 T91
3	Inner horizontal	20.0 m	12Cr2MoWVTiB
4	Upper horizontal	12.0 m	12Cr2MoWVTiB
5	Outlet casing	2.5 m	12Cr1MoV

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2. Installing the app

1. Obtain `app-debug.apk` (built from `./gradlew :app:assembleDebug`, located in `app/build/outputs/apk/debug/`).

1. Copy it to your Android device (USB, cloud, or direct download).
2. On the device, open the APK. Android will ask to allow installation from that source ("Install unknown apps") — allow it once.

1. Tap **Install**, then open **Platen Superheater Transient**.

No special permissions are required. The app works fully offline — the IF97 property library is compiled into the app.

3. How the simulation works — the big picture

The platen is **not** modeled as one lump. It is divided into the **five physical sections** listed above, and each section carries its own state:

- **Metal temperature** `T_m` — the tube-wall metal lump (dominant inertia)
- **Steam temperature** `T_s` — the steam inside that section

Each timestep the model:

1. Reads the current inputs (steam flow, spray flow/temperature, burners, pressure, steam temperature) — including any scenario events due at this time.

1. If spray is active, performs the **mixing calculation** (see §5): enthalpy balance → mixed enthalpy → mixed temperature via the IF97 backward equation T(P,h).

1. For every segment, evaluates properties (ρ, μ, cp, k) from IF97 at the current steam state, computes velocity → Re → Pr → Nu → h_i, then U from the cylindrical resistance chain.

1. Computes the per-segment heat flows and integrates both energy balances with the **RK4** method (4th-order Runge–Kutta).

1. Passes each segment's outlet steam temperature as the next segment's inlet temperature (segment 1 receives the post-spray mixed temperature).

1. Records everything for the charts.

Why the metal matters: the tube metal holds ~36 MJ/K — thousands of times more thermal capacity than the steam in the tubes. When burners step up, the metal heats first and the steam follows; that lag is exactly what the model reproduces.

Flow distribution assumption (explicit): total platen steam flow splits equally across all 43 panels; within a panel the 4 tubes share equally. So one tube carries `ṁ_total / 172`. At full load (277.8 kg/s) and 167 bar / 520 °C (ρ ≈ 52 kg/m^3) this gives ≈ 47 m/s tube velocity and Re ≈ 5·10^5 — physically correct for superheated steam.

4. The Simulate tab — step by step

4.1 Initial state fields

Field	Meaning	Default
Steam flow t/h	Total platen steam flow at t = 0	100
Pressure bar	Steam pressure at the platen inlet (absolute)	100
Steam T °C	Steam temperature entering the platen at t = 0	400
Spray T °C (<250)	Spray-water temperature — must be below 250 °C	230
Burners on	Number of burners firing at t = 0	2
Duration min	Simulation length	15

4.2 Scenario event fields (step at t = 300 s)

One built-in event time (300 s) is provided for quick what-if tests:

Field	Meaning
Flow → t/h	New steam flow applied at t = 300 s (blank/0 = no change)
Spray → t/h	New spray flow applied at t = 300 s (0 = spray off)
Burners →	New burner count applied at t = 300 s

Leave all three empty to simulate the initial state alone (useful for seeing the natural heat-up toward equilibrium).

4.3 Running

Tap **RUN SIMULATION**. The engine integrates with dt = 1 s. Results appear below: summary numbers first, then four charts, then the calculation-details panel.

If anything is wrong with the inputs (e.g. spray hotter than 250 °C, non-physical pressure), a red error card appears with an engineering-readable explanation instead of a crash.

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5. Spray water rules (important)

The app **enforces the plant rule** for spray water:

> **Spray water is always subcooled and below 250 °C — in any situation.**

Concretely, the engine checks, before every mixing calculation:

1. **Temperature limit:** spray temperature must be **< 250 °C**.
2. **Subcooling:** spray temperature must be **below the saturation temperature Tsat**

at the mixing pressure. At 167 bar, Tsat ≈ 350 °C, so the 250 °C limit is the binding one there; at 20 bar, Tsat ≈ 212 °C, so subcooling binds first.

If either check fails, the simulation refuses with a typed error explaining exactly which condition failed and by how much — for example:

```
Spray temperature 262.0 °C exceeds the 250 °C plant limit
Spray water must be subcooled: T=220.0 °C >= Tsat=212.4 °C at 20.00 bar
```

Why: desuperheating sprays must evaporate completely in the steam stream. Water that is saturated or boiling at the injection pressure flashes rather than mixing, which the single-zone mixing model does not represent. The rule keeps every result in the physically valid region.

Spray temperature as an event: you can also change spray temperature during a scenario (`SPRAY_TEMP_K` events); the same rule is enforced at every event.

The mixing calculation (what happens when spray is on)

Four distinct states are always kept separate:

1. **Upstream steam** (pre-spray): P, T, h from IF97
2. **Spray water**: h from IF97 Region 1 at the injection pressure and spray temperature
3. **Mixed state** (post-spray, platen inlet): enthalpy balance → T(P,h)
4. **Platen outlet**: computed by the segment chain

```
h_mix = (ṁ_steam·h_steam + ṁ_spray·h_spray) / (ṁ_steam + ṁ_spray)
T_mix = T(P, h_mix)          ← IF97 Region 2 backward equation
```

The cooling power attributable to the spray is `Q_spray = ṁ_spray·(h_steam − h_mix)`.

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6. Scenario events

Internally every input is a **time series** built from events. The built-in UI exposes one event time; the engine supports arbitrary lists (the full scenario editor ships in a coming release):

Event kind	Units	Default behavior
Steam flow	kg/s (UI: t/h)	Step at event time
Spray flow	kg/s (UI: t/h)	Step (or ramp if spray ramp configured)
Spray temperature	K (UI: °C)	Step; must stay < 250 °C
Burners firing	count	**Ramped** over 60 s (configurable), not stepped
Firing fraction	0–1	Step
Steam pressure	Pa (UI: bar)	Step
Steam temperature	K (UI: °C)	Step

The example scenario from the specification is exactly the default UI event set:

t (s)	Event
0	Steam flow = 100 t/h, 2 burners on, spray = 0
300	Steam flow → 150 t/h
300	Spray 0 → 5 t/h
300	Burners 2 → 3

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7. Reading the results and charts

Summary numbers (at end of simulation)

Row	Meaning
Outlet T (start → end)	Steam temperature after the last segment, start vs end
Metal T avg (end)	Average metal temperature across the 5 segments
Mixed T (end)	Post-spray mixed temperature (equals steam T if no spray)
h_i (end)	Internal convection coefficient, W/m²K
U (end)	Overall heat-transfer coefficient, W/m²K
Re (end)	Reynolds number in a tube
Q absorbed (end)	Heat picked up by the steam across all segments, MW

Charts

Each chart plots the quantity against time; the axis label shows the value range

covered:

1. **Outlet temperature** — the headline result. Watch the step response after events.
2. **Metal temperature** — leads the outlet; shows the thermal-inertia delay.
3. **h_i vs time** — internal convection responding to flow/state changes.
4. **U vs time** — overall coefficient; dominated by the weakest resistance.

How to interpret a step response: after a burner step the metal temperature begins rising immediately, and the outlet follows after a short dead time; after a spray step the outlet begins falling almost immediately (the spray acts at the inlet), settling toward a new steady value.

8. Show calculation — full transparency

Every result must be traceable. Tap **Show calculation details** at the bottom of the results to see, for the end-of-simulation state:

Line	Source
P, T_out	Current steam state at the outlet
ρ	IF97 density at (P, T_out)
cp	IF97 isobaric heat capacity
μ	IF97 dynamic viscosity
k	IF97 thermal conductivity
Pr	IF97 Prandtl number
Re	ρ·v·D_i/μ from the flow model
Nu	0.023·Re^0.8·Pr^0.4 (Dittus–Boelter)
h_i = Nu·k/D_i	internal convection
U	cylindrical resistance chain

This is a core requirement of the tool (engineering traceability), not a convenience feature.

9. The Validation tab — model comparison

The app ships **three model levels** and runs them side by side on the same scenario:

Model	Description	Purpose
Constant-U	Single block, fixed U (800 W/m²K)	Plant-YAML reference behavior
Dynamic h_i	Single block, h_i from the Re→Pr→Nu chain each step	Isolates the effect of dynamic heat transfer
Segmented (RK4)	Full 5-segment transient model	The production model

Tap **Run comparison** to see the end-of-run outlet temperature of all three on the reference scenario (100→150 t/h and 2→3 burners at t = 300 s).

The 800 vs ~170 W/m²K discrepancy is surfaced, not hidden. Plant YAML data gives U ≈ 800 W/m²K at low load, while the Dittus–Boelter chain gives ~170 W/m²K at full load. Both paths are provided as separate, clearly labeled modes — the app never silently blends them. Resolving the gap is a calibration exercise (next section), not something the app decides for you.

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10. Calibration against plant data

The **Fit parameters** workflow lets you calibrate two effective parameters against measured plant data:

1. Export a window of historian data containing the **outlet temperature** sampled evenly (the simulation grid is 1 s × 900 s; resample your data to match, or paste a representative series of at least 10 points).

1. Paste the values (°C), comma- or newline-separated, into the text field.
2. Tap **Fit parameters**.

The fitter grid-searches:

- **F_platen** (fraction of furnace release reaching the platen) in 0.05…0.5
- **h_o** (external/furnace-side convection) in 20…500 W/m²K

and reports fit quality:

Metric	Meaning
MAE	Mean absolute error, K
RMSE	Root-mean-square error, K
Max error	Worst instantaneous error, K
Bias	Mean signed error, K (model too hot or too cold overall)
R^2	Fraction of variance explained

Fitted values are labeled empirical and kept separate from the physical parameters — the app never overwrites physics with fitted numbers. Use the fitted F_platen/h_o as the configuration for subsequent simulation runs.

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11. The physics inside — equations used

11.1 Property engine (IAPWS-IF97)

All thermophysical properties come from IAPWS-IF97:

- **Region 1** — compressed/subcooled water (feedwater, spray water)
- **Region 2** — superheated steam (main flow, and T(P,h) backward equation)
- **Region 4** — saturation line (Tsat for the spray subcooling check)

Verified in the test suite against the official IAPWS tables, e.g.:

Quantity	State	Official value
h	3 MPa, 300 K	115.331 kJ/kg
h	3 MPa, 500 K	975.542 kJ/kg
h	30 MPa, 700 K	2631.495 kJ/kg
μ	0.1 MPa, 298.15 K	0.890022551 mPa·s
k	0.1 MPa, 298.15 K	0.607509806 W/m·K
Tsat	16.7 MPa	623.988 K

11.2 Flow

```
A_flow   = π·D_i²/4                  D_i = 29 mm
ṁ_tube   = ṁ_total / (43·4)
v        = ṁ_tube / (ρ·A_flow)
Re       = ρ·v·D_i/μ
```

11.3 Heat transfer

```
Pr   = cp·μ/k
Nu   = 0.023·Re^0.8·Pr^0.4          (heating; n = 0.3 for cooling)
h_i = Nu·k/D_i
```

Dittus–Boelter validity (Pr 0.7–160, Re > 10 000) is checked and warned, never

silently applied outside its range.

11.4 Overall U (cylindrical, per outer area)

```
1/(U·A_o) = 1/(h_i·A_i) + ln(D_o/D_i)/(2π·k_wall·L) + R_fouling + 1/(h_o·A_o)
```

All four resistances are exposed; the dominant one is identified in the results.

11.5 Furnace heat input

```
Q_platen = n_burners · 40 MW · firing_fraction · F_platen
```

F_platen default 0.15. Burner changes ramp over 60 s.

11.6 Per-segment energy balances

```
C_metal · dT_m/dt = Q_furnace,seg − h_i·A_i·(T_m − T_s) − Q_loss
C_steam · dT_s/dt = ṁ·cp·(T_in − T_s) + h_i·A_i·(T_m − T_s)
```

The steam balance is integrated in exponential-relaxation form with `τ = max(C_steam/(ṁ·cp + h_i·A_i), Δt)` — the same steady state, stable at any segment size (the short casing segments hold very little steam).

11.7 Solver

RK4, default Δt = 1 s (configurable 0.1–2 s). Verified by the timestep-halving convergence test: dt = 2 s and dt = 1 s agree within 5 K on the reference scenario.

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12. Units

Context	Units
UI display	bar, °C, t/h, MW, W/m²K
Engine internals	SI only: Pa, K, kg/s, J/kg, W, m, kg/m³, J/kg·K, Pa·s, W/m·K

Conversions happen exactly once, at the UI boundary. Internally nothing is ever mixed.

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13. Error messages and what they mean

Message	Cause	What to do
`Spray temperature X °C exceeds the 250 °C plant limit`	Spray T ≥ 250 °C	Set spray temperature below 250 °C
`Spray water must be subcooled: T=… >= Tsat=… at … bar`	Spray T ≥ Tsat at mixing pressure	Lower spray temperature below Tsat
`IF97 state out of range: …`	Pressure/temperature outside IF97 validity (e.g. > 100 MPa, < 0 °C)	Bring inputs into the physical range
`Steam flow must be positive`	Zero/negative flow entered	Enter a positive flow
`Simulation error at t=…`	A state went non-physical mid-run	Reduce event magnitude, check initial state

All errors are typed results — the app never crashes on bad input; it explains.

14. FAQ and troubleshooting

Q: The outlet temperature keeps rising the whole run — is that wrong?

A: Probably not. The initial metal temperature equals the initial steam temperature; if the burner heat input exceeds what the flow carries away, the system is heating toward a higher equilibrium and 15 minutes is not enough to settle. Extend the duration or compare spray-on vs spray-off runs (the Validation comparison does this).

Q: Why is tube velocity ~47 m/s at full load? Isn't that high?

A: Superheated steam at 167 bar/520 °C has density ≈ 52 kg/m^3 — a tenth of water. High velocity at full load is physically correct; Re ≈ $5{\cdot}10^5$ is comfortably turbulent.

Q: Why does the metal respond before the outlet?

A: The furnace heats the metal directly; the steam only receives heat through the metal. That cascade delay is the dominant transient behavior of a superheater.

Q: Can I set spray to 300 °C to test the limit?

A: The engine will refuse — that is the plant rule working as intended. Spray water is always subcooled and below 250 °C in this plant.

Q: What pressure should I enter — absolute or gauge?

A: Absolute (bar a). IF97 works in absolute pressure.

Q: Where does the IF97 library come from?

A: `com.hummeling.if97` v2.1.0 by Hummeling Engineering BV, vendored in source form under the LGPL (license file included in the source tree).

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15. Building the app from source

```
# Requirements: JDK 17, Android SDK (see local.properties), internet for Gradle deps
git clone https://github.com/mohsenRahimi1708/SteamMixerCalculator.git
cd SteamMixerCalculator   # checkout unit4-platen-transient
# the app lives in the platen-superheater/ folder of that branch
./gradlew :app:testDebugUnitTest   # 17 tests, all green
./gradlew :app:assembleDebug       # APK at app/build/outputs/apk/debug/app-debug.apk
```

Architecture summary:

```
UI (Compose) → Screens → PlatenSimulator → SteamProperties (IF97)
                              ↓
              FlowModel / HeatTransfer / OverallU
                              ↓
              BurnerModel / SprayModel / ScenarioState
```

The engine package is pure Kotlin/JVM (no Android dependencies) and fully unit-tested; the property engine sits behind an interface so an alternative implementation (e.g. CoolProp) can be swapped in without touching the model.

Document version 1.0.0 — generated with the application source. For the full engineering derivation of every equation, see `ENGINEERING.md` in the repository.